

Ans to the question no 1

(a) The average velocity over the time interval $[1, 3]$ is,

$$V_{\text{avg}} = \frac{f(3) - f(1)}{3 - 1} = \frac{1 - (-1)}{2} = 1 \text{ m/s} \quad (\text{ans})$$

The average velocity over the time interval $[5, 8]$ is,

$$V_{\text{avg}} = \frac{f(8) - f(5)}{8 - 5} = \frac{(-2) - 1}{3} = -1 \text{ m/s} \quad (\text{ans})$$

(b) At $t = 1$ and at $t = [3, 5]$ (horizontal tangent line)

(c) Increasing intervals : $(1, 3)$ and $(8, 9)$

Decreasing intervals : $(0, 1)$ and $(5, 8)$

Stand-still interval : $(3, 5)$

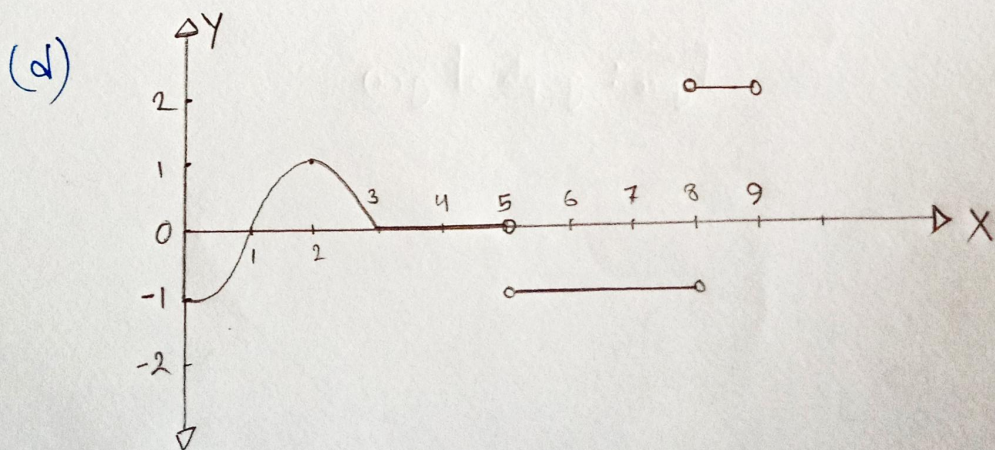


Figure : velocity graph

Ans to the question no 2

(a) $g'(x)$ is positive when $x = -3$

$g'(x)$ is zero when $x = -1, 3, 6$

$g'(x)$ is negative when $x = 1$

(b) $g'(x) = 0$ at $x = -1, 3, 6$

because the slope of $g(x)$ is zero when $x = -1, 3, 6$.

(c)

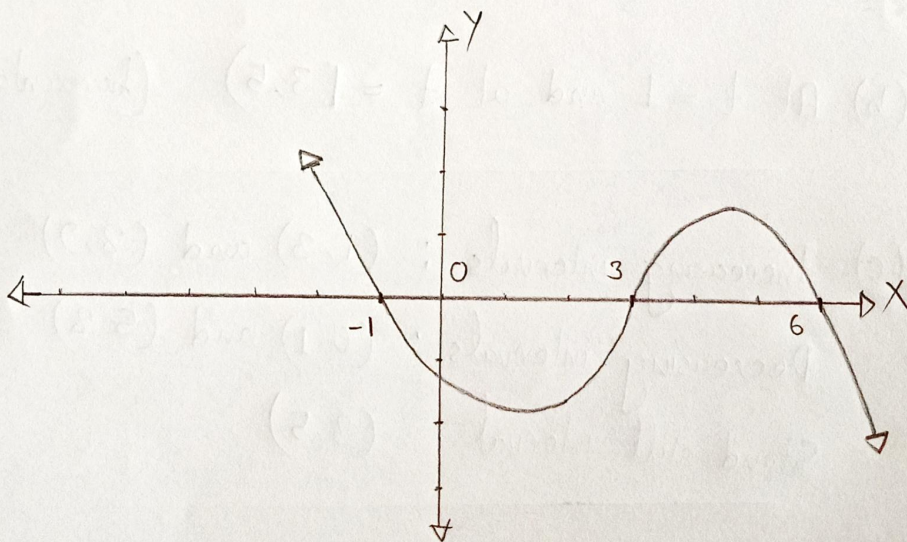


figure: graph of $g'(x)$

Ans to the question no 3

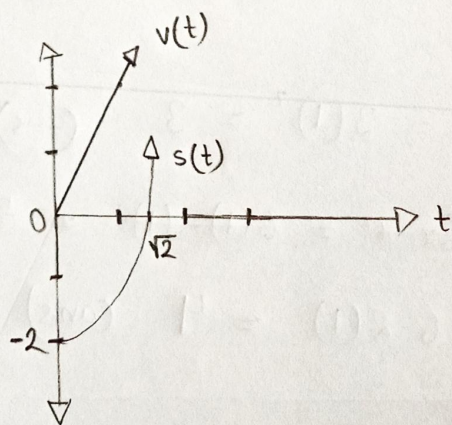
(a) For the time interval. $t = [2, 3]$,

$$V_{avg} = \frac{s(3) - s(2)}{3 - 2} = \frac{3^2 - 2 - (2^2 - 2)}{1} = 5 \text{ m/s (ans)}$$

(b) The particle's instantaneous velocity at time, $t = 2$ s is,

$$\begin{aligned} V_{ins} &= \lim_{h \rightarrow 0} \frac{s(2+h) - s(2)}{h} = \lim_{h \rightarrow 0} \frac{(2+h)^2 - 2 - (2^2 - 2)}{h} = \lim_{h \rightarrow 0} \frac{4h + h^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{h(4+h)}{h} = \lim_{h \rightarrow 0} 4 + h = 4 \text{ m/s (ans)} \end{aligned}$$

(c)



$$s(t) = t^2 - 2$$

$$v(t) = 2t$$

$$t \geq 0 \text{ s}$$

figure: $s(t)$ and $v(t)$

Ans to the question no. 04

(i) for (a) $f(x) = x^3 + 3$,

$$\begin{aligned} m_{\text{int}} &= \lim_{x_0 \rightarrow x} \frac{f(x_0) - f(x)}{x_0 - x} = \lim_{x_0 \rightarrow x} \frac{x_0^3 + 3 - x^3 - 3}{x_0 - x} = \lim_{x_0 \rightarrow x} \frac{x_0^3 - x^3}{x_0 - x} \\ &= \lim_{x_0 \rightarrow x} \frac{(x_0 - x)(x^2 + xx_0 + x_0^2)}{(x_0 - x)} = x^2 + xx + x^2 = 3x^2 \quad (\text{ans}) \end{aligned}$$

for (b) $g(x) = 6x - x^2$

$$\begin{aligned} m_{\text{int}} &= \lim_{x_0 \rightarrow x} \frac{f(x_0) - f(x)}{x_0 - x} = \lim_{x_0 \rightarrow x} \frac{6x_0 - x_0^2 - 6x + x^2}{x_0 - x} = \lim_{x_0 \rightarrow x} \frac{6(x_0 - x) - (x_0^2 - x^2)}{(x_0 - x)} \\ &= \lim_{x_0 \rightarrow x} \frac{6 - (x_0 + x)}{1} = 6 - 2x \quad (\text{ans}) \end{aligned}$$

(ii) from i we got,

$$m_1 = f'(1) = 3x^2 = 3(1)^2 = 3 \quad (\text{ans})$$

$$\begin{aligned} \text{and } m_2 &= g'(1) = 6x - x^2 = 6(1) - (1)^2 = 5 \quad (\text{ans}) \\ &= 6 - 2x = 6 - 2(1) = 4 \quad (\text{ans}) \end{aligned}$$

$$(iii) f(1) = (1)^3 + 3 = 4$$

$\therefore$ at $x=1$ we get point $(1, 4)$ on $f(x)$

$$g(1) = 6(1) - (1)^2 = 5$$

$\therefore$ at $x=1$ we get point $(1, 5)$ on $g(x)$

So tangent line equation of $f(x)$ at $x=1$ is,

$$y - 4 = m_1(x-1) = 3(x-1) = 3x-3$$

$$\therefore y = 3x+1 \quad (\text{ans})$$

and tangent line equation of $g(x)$ at $x=1$ is,

$$y - 5 = m_2(x-1) = 4x - 4 \times 1 = 4x - 4$$

$$\therefore y = 4x+1 \quad (\text{ans})$$

(iv) At $x = \pm 1$; we get points $(-1, 2)$ & $(1, 4)$ on $f(x)$

$$f(x) \text{ secant slope at } [-1, 1], m_{\text{sec}} = \frac{4-2}{1-(-1)} = 1$$

$\therefore f(x)$ secant line equation at interval $[-1, 1]$,

$$y - 2 = 1(x - (-1)) = x + 1$$

$$\therefore y = x + 3 \quad (\text{ans})$$

At $x = \pm 1$; we get points $(1, 5)$ & $(-1, -7)$ on $g(x)$

$$g(x) \text{ secant line slope at } [-1, 1], m_{\text{sec}} = \frac{-7-5}{-1-1} = \frac{-12}{-2} = 6$$

$\therefore g(x)$ secant line equation at interval $[-1, 1]$,

$$y - 5 = 6(x - 1) = 6x - 6$$

$$\therefore y = 6x - 1 \quad (\text{ans})$$

(v)

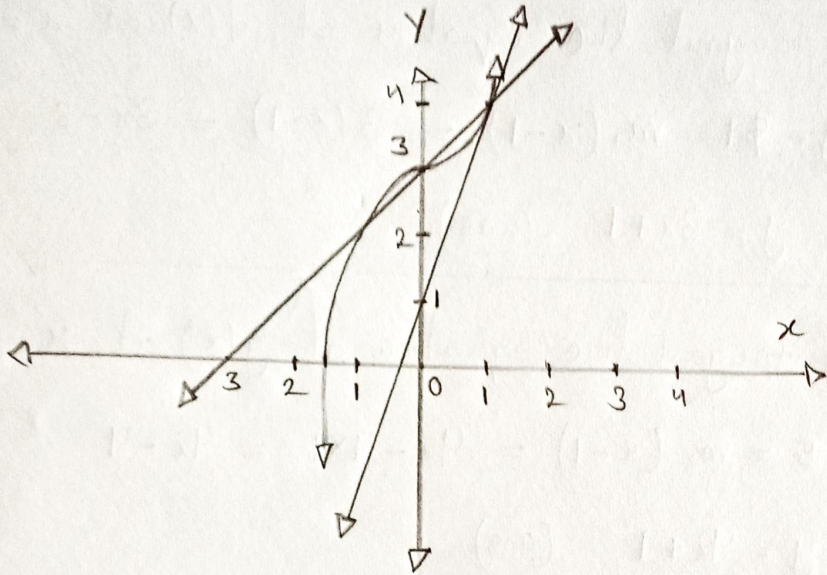


figure: $f(x) = x^3 + 3$

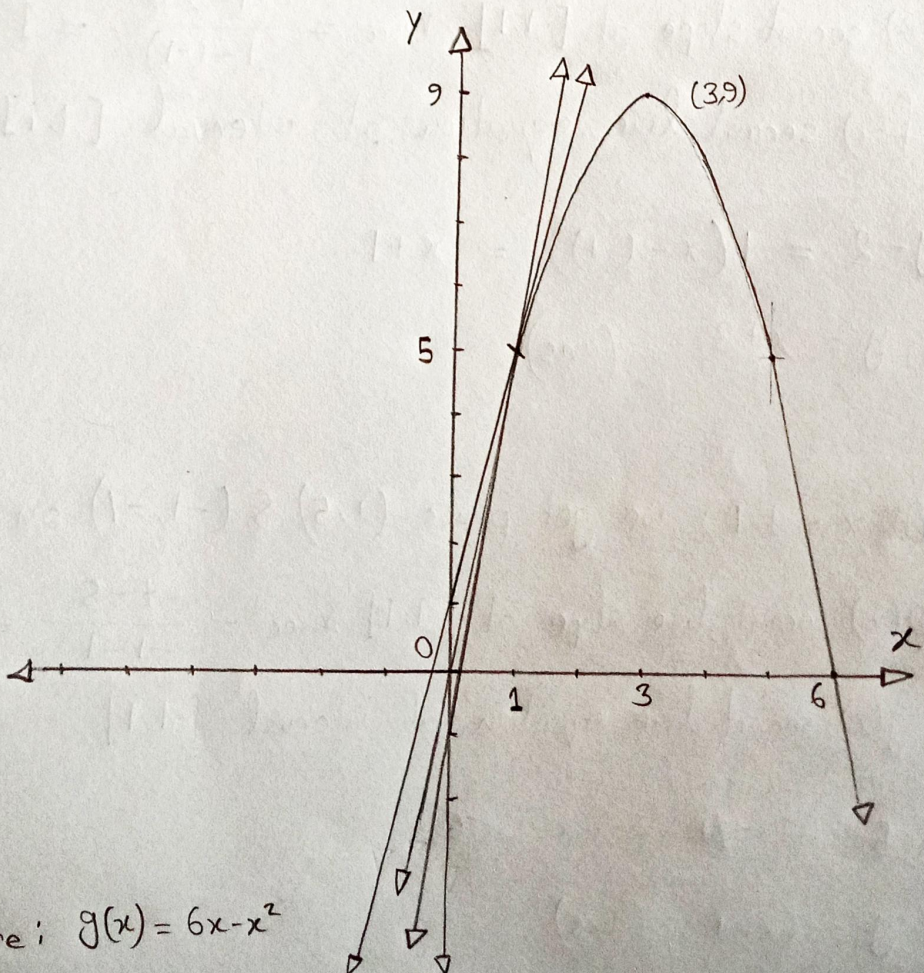


figure: $g(x) = 6x - x^2$

Ans to the question no 5

(i) for $f(x) = -3x^2$

$$\begin{aligned} f'(x) &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0} \frac{-3(x+h)^2 + 3x^2}{h} \\ &= \lim_{h \rightarrow 0} \frac{-3x^2 - 6xh - 3h^2 + 3x^2}{h} = \lim_{h \rightarrow 0} \frac{-h(6x + 3h)}{h} \\ &= \lim_{h \rightarrow 0} -(6x + 3h) = -6x \quad (\text{ans}) \end{aligned}$$

for $g(x) = x^3 - 4$

$$\begin{aligned} g'(x) &= \lim_{h \rightarrow 0} \frac{g(x+h) - g(x)}{h} = \lim_{h \rightarrow 0} \frac{(x+h)^3 - 4 - x^3 + 4}{h} \\ &= \lim_{h \rightarrow 0} \frac{x^3 + 3x^2h + 3xh^2 + h^3 - x^3}{h} = \lim_{h \rightarrow 0} \frac{h(3x^2 + 3xh + h^2)}{h} \\ &= 3x^2 + 3x(0) + (0)^2 = 3x^2 \quad (\text{ans}) \end{aligned}$$

(ii) At $x = (-1)$, $f(-1) = -3(-1)^2 = -3$

$$\text{and } m = f'(-1) = -6(-1) = 6$$

$\therefore$ At $x = (-1)$, equation of the tangent line to $f(x)$ is,

$$y + 3 = 6(x + 1) = 6x + 6$$

$$\Rightarrow y = 6x + 3 \quad (\text{ans})$$

$$\text{At } x = (-1), g(-1) = (-1)^3 - 4 = -5$$

$$\text{and } m = g'(-1) = 3(-1)^2 = 3$$

$\therefore$ At $x = (-1)$, equation of the tangent line to $g(x)$ is,

$$y + 5 = 3(x + 1) = 3x + 3$$

$$\Rightarrow y = 3x - 2 \quad (\text{ans})$$

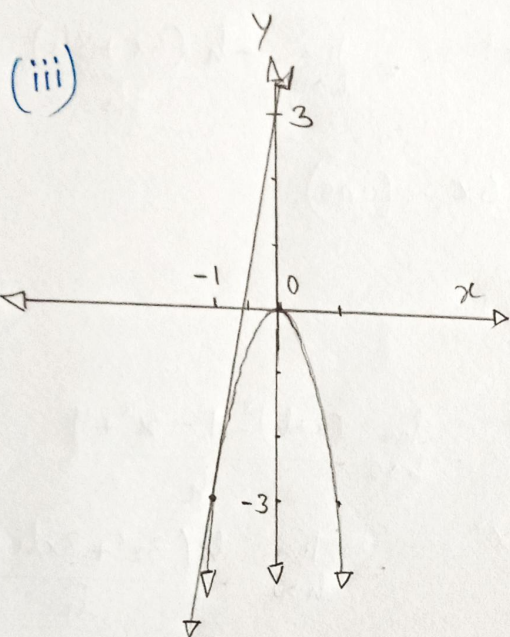


figure $f(x)$

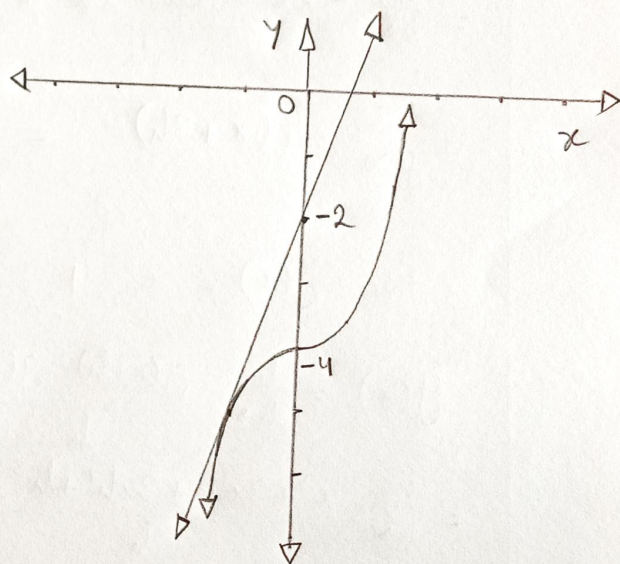


figure $g(x)$

Ans to the question no 6

(a)

Given, $s(t) = t^4 - 2t^3 + t^2 - t$

$$\therefore \text{Velocity } v(t) = 4t^3 - 6t^2 + 2t - 1 = s'(t)$$

$$\therefore \text{Acceleration } a(t) = 12t^2 - 12t + 2 = s''(t)$$

(b) Given, $t = 1s$

$$\therefore \text{Acceleration, } s''(1) = a(1) = 12(1)^2 - 12(1) + 2 \\ = 2 \text{ m/s}^2 \text{ (ans)}$$

Ans to the question no 7

(a) Here, $\lim_{x \rightarrow 0^+} 2 - |x| = \lim_{x \rightarrow 0^-} 2 - |x| = f(0) = 2.$

So at $x = 0$, $f(x)$ is continuous.

$$\lim_{h \rightarrow 0} f(x) = \begin{cases} 2-x, & x \geq 0 \\ 2+x, & x < 0 \end{cases}$$

$$\lim_{h \rightarrow 0^-} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0^-} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0^-} \frac{2+h-2}{h} \\ = +1$$

$$\lim_{h \rightarrow 0^+} \frac{f(x+h) - f(x)}{h} = \lim_{h \rightarrow 0^+} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0^+} \frac{2-h-2}{h} \\ = -1$$

$$\text{As } \lim_{h \rightarrow 0^-} \frac{f(x+h) - f(x)}{h} \neq \lim_{h \rightarrow 0^+} \frac{f(x+h) - f(x)}{h},$$

$f'(0)$ does not exist or $f(x)$ is not differentiable at $x = 0$. (ans)

(b) Here, $f(x) = \begin{cases} 2-x, & x > 0 \\ 2+x, & x < 0 \end{cases}$

$$\text{So } f'(x) = \begin{cases} -1, & x > 0 \\ 1, & x < 0 \end{cases} \quad (\text{ans})$$

Ans to the question no 8

If a function f is not continuous at x , then f is not differentiable at x .

The graph $f(x)$ is not continuous at $x = -8, 0, 3$, so $f(x)$ is not differentiable at these values of x .

Also a function f is not differentiable at values of x , where graph of f has corner points and points of verticle tangency.

The graph $f(x)$ has corner points and verticle tangency when $x = -4, 2$. So $f(x)$ is also not differentiable at these values of x .

Ans to the question no 9

a $\rightarrow$ II

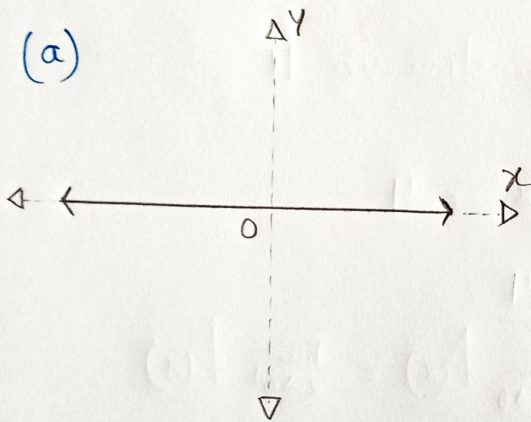
b $\rightarrow$ IV

c $\rightarrow$ I

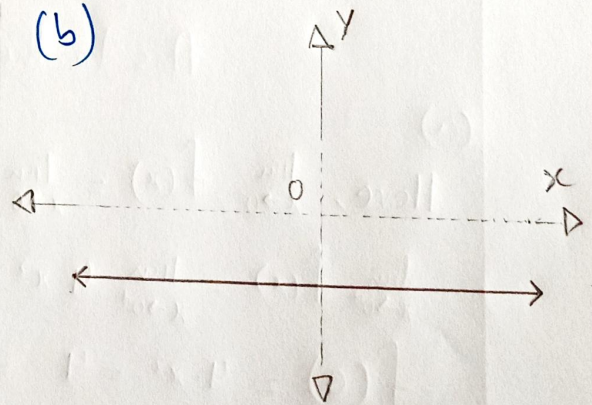
d $\rightarrow$ III

Ans to the question no 10

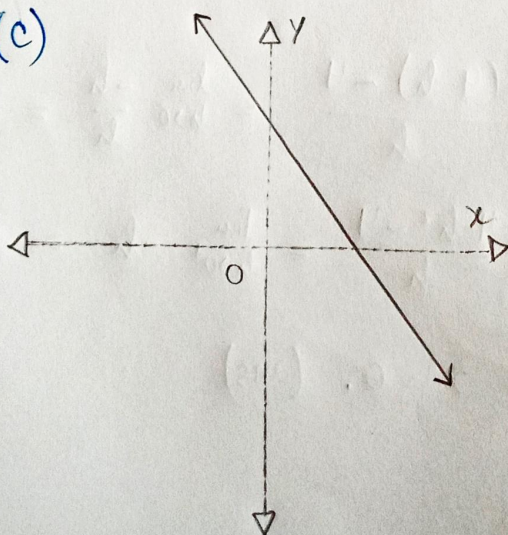
(a)



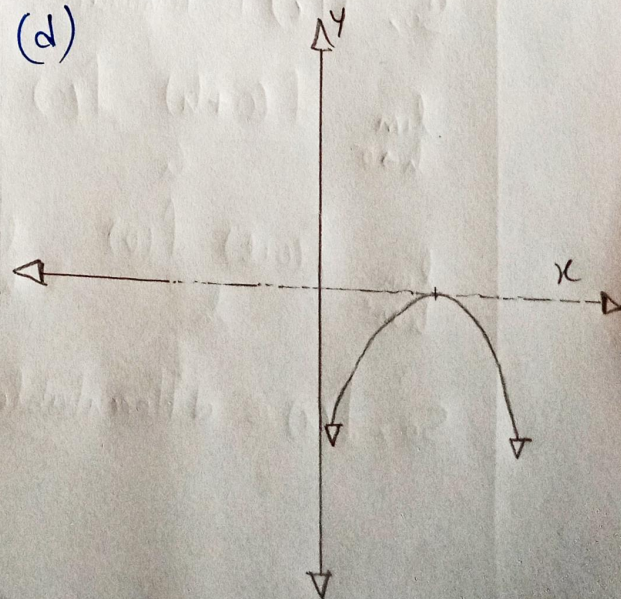
(b)



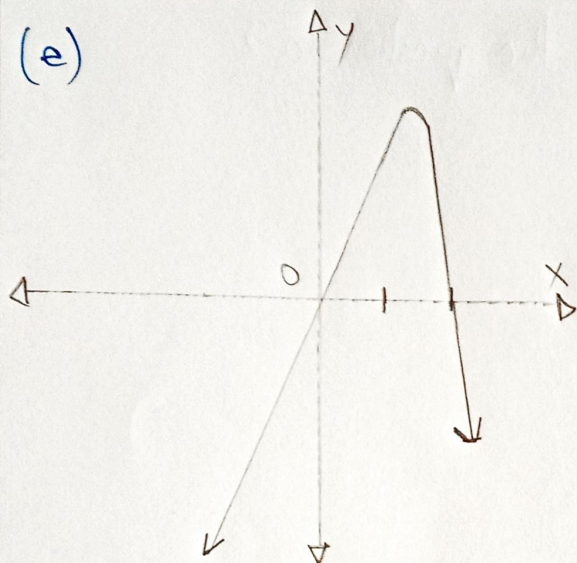
(c)



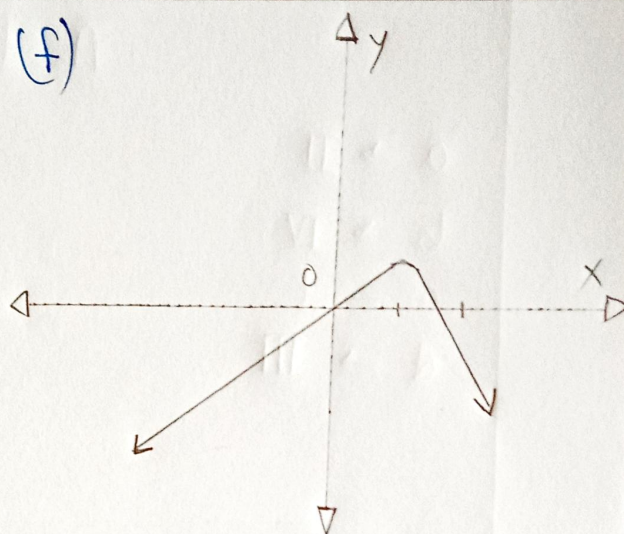
(d)



(e)



(f)



Ans to the question no 11

(a)

$$\text{Here, } \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^-} 4 - x = 4$$

$$\lim_{x \rightarrow 0^+} f(x) = \lim_{x \rightarrow 0^+} 4 - x^2 = 4$$

$$f(0) = 4 - x^2 = 4 = \lim_{x \rightarrow 0^-} f(x) = \lim_{x \rightarrow 0^+} f(x)$$

So, $f(x)$ is continuous at $x=0$. (ans)

$$\lim_{h \rightarrow 0^-} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0^-} \frac{(4-h) - 4}{h} = \lim_{h \rightarrow 0^-} \frac{-h}{h} = -1$$

$$\lim_{h \rightarrow 0^+} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0^+} \frac{4-h^2 - 4}{h} = \lim_{h \rightarrow 0^+} -h = 0$$

So, $f(x)$ is differentiable at $x=0$. (ans)

(b) Here, $f(x) = \begin{cases} 4-x & \text{if } x < 0 \\ 4-x^2 & \text{if } x \geq 0 \end{cases}$ $f'(x) = \begin{cases} -1 & \text{if } x < 0 \\ -2x & \text{if } x > 0 \end{cases}$

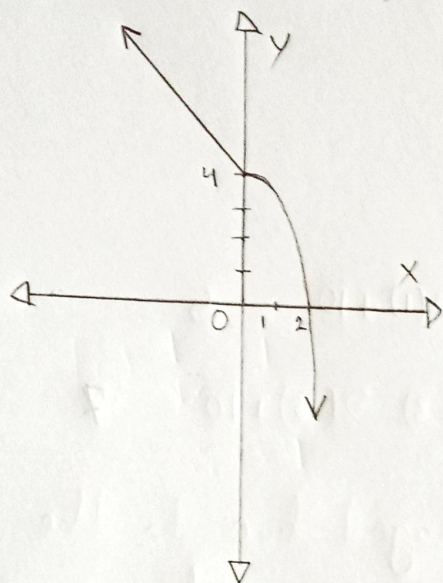


figure: $f(x)$

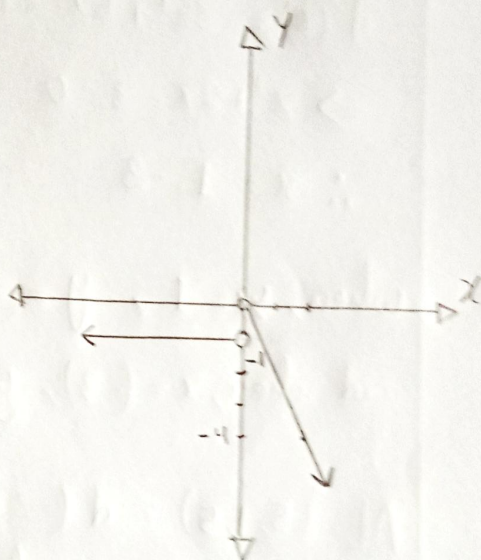


figure: $f'(x)$

Ans to the question no 12

(a) Curve $y = x^3 - 3x^2 + 7$ will be parallel to the line $y = 9x - 1$, where derivative of the curve is equal to slope of line which is 9.

$$y_1 = 3x^2 - 6x = 9$$

$$\Rightarrow x^2 - 2x - 3 = 0$$

$$\Rightarrow x^2 - 3x + x - 3 = 0$$

$$\Rightarrow (x-3)(x+1) = 0$$

$$\therefore x = 3, (-1)$$

$$\text{when, } x = 3, y = 3^3 - 3 \cdot 3^2 + 7 = 7$$

$$\text{and when, } x = (-1), y = (-1)^3 - 3(-1)^2 + 7 = 3$$

$\therefore$ So at point $(3, 7)$ & $(-1, 3)$ the curves tangent line is parallel to the line $y = 9x - 1$. (ans)

(b) Tangent line is horizontal at points where $f'(x) = 0$.

$$f'(x) = 3x^2 + 6x - 9 = 0$$

$$\Rightarrow x^2 + 2x - 3 = 0$$

$$\therefore x = 1, -3$$

$$\text{when, } x = 1, y = 1^3 + 3(1)^2 - 9(1) + 10 = 5$$

$$\text{and when, } x = (-3), y = (-3)^3 + 3(-3) - 9(-3) + 10 = 37$$

At $(1, 5)$ and $(-3, 37)$, the tangent line to the curve $f(x)$ has horizontal tangent is horizontal. (ans)

(c) On curve $y = 2\sqrt{x}$, tangent line is perpendicular to

$$y = 3 - 2x \quad \text{where } f'(x) = \frac{1}{2}$$

$$\therefore f'(x) = \frac{1}{\sqrt{x}} = \frac{1}{2}$$

$$\Rightarrow x = 4$$

$$\text{when } x = 4, y = 2\sqrt{4} = 4$$

So, at $(4, 4)$ the tangent line of $y = 2\sqrt{x}$ is perpendicular to the line $y = 3 - 2x$. (ans)

Ans to the question no 13

$$(a) \quad h(x) = 3f(x) - 2g(x) + f(x)g(x)$$

$$\Rightarrow h'(x) = 3f'(x) - 2g'(x) + f'(x)g(x) + f(x)g'(x)$$

$$\Rightarrow h'(3) = 3f'(3) - 2g'(3) + f'(3)g(3) + f(3)g'(3)$$

$$= 3 \times 6 - 2(-2) + 6(-5) + 7(-2)$$

$$= -22 \quad (\text{ans})$$

$$(b) \quad h(x) = \frac{f(x)}{\ln x}$$

$$\Rightarrow h'(x) = \frac{f'(x)\ln x - f(x)\frac{1}{x}}{(\ln x)^2}$$

$$\Rightarrow h'(3) = \frac{f'(3)\ln 3 - f(3)\frac{1}{3}}{(\ln 3)^2} = \frac{6\ln 3 - 7/3}{(\ln 3)^2} \quad (\text{ans})$$

$$(c) \quad h(x) = e^x g(x)$$

$$\Rightarrow h'(x) = e^x g(x) + e^x g'(x)$$

$$\Rightarrow h'(3) = e^3 g(3) + e^3 g'(3) = e^3 (-5-2) = -7e^3$$

$$(d) \quad h(x) = \frac{1+\sqrt{x}f(x)}{x^2+g(x)}$$

$$\Rightarrow h'(x) = \frac{\{x^2+g(x)\}\left\{\frac{f(x)}{2\sqrt{x}} + \sqrt{x}f'(x)\right\} - (1+\sqrt{x}f(x))\{2x+g'(x)\}}{\{x^2+g(x)\}^2}$$

$$\Rightarrow h'(3) = \frac{4\left(\frac{7}{2\sqrt{3}} + 6\sqrt{3}\right) - (1+7\sqrt{3})^4}{16} = \frac{-6+\sqrt{3}}{24} \quad (\text{ans})$$

Ans to the question no 14

$$(a) f(x) = \sec^3(x^5) - x^2 \cos 3x - 5$$

$$\begin{aligned} f'(x) &= 3\sec^2 x^5 \cdot \frac{d}{dx} \sec x^5 - \left(\cos 3x \frac{d}{dx} x^2 + x^2 \frac{d}{dx} \cos 3x \right) - \frac{d}{dx}(5) \\ &= 3\sec^2 x^5 \cdot \sec x^5 \cdot \tan x^5 \cdot \frac{d}{dx} x^5 - (2x \cos 3x - 3x^2 \sin 3x) - 0 \\ &= 3\sec^2 x^5 \cdot \sec x^5 \cdot \tan x^5 \cdot 5x^4 - 2x \cos 3x + 3x^2 \sin 3x \quad (\text{ans}) \end{aligned}$$

$$\begin{aligned} (b) \frac{dt}{dx} &= \frac{dt}{dy} \cdot \frac{dy}{dx} = \frac{\frac{d}{dy}(\sin(y^2+1))}{\sin(y^2+1)} \cdot (-1) \sin(x^2 e^x) \cdot \frac{d}{dx}(x^2 e^x) \\ &= \frac{\cos(y^2+1)}{\sin(y^2+1)} \cdot \frac{d}{dy}(y^2+1) \cdot (-1) \sin(x^2 e^x) (2x e^x + x^2 e^x) \\ &= -2y \cot(y^2+1) \cdot \sin(x^2 e^x) \cdot e^x x (2+x) \\ &= -2 \cos(x^2 e^x) \cot(\cos^2(x^2 e^x)+1) \cdot \sin(x^2 e^x) \cdot e^x x (2+x) \\ &= -e^x x \cot(\cos^2(x^2 e^x)+1) \sin(2e^x x^2) (2+x) \quad (\text{ans}) \end{aligned}$$

Ans to the question no 15

$$(a) \frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$$

$$(a) \text{ let } u = \cos x, \text{ then } y = \ln(u)$$

$$\begin{aligned} \therefore \frac{dy}{dx} &= \frac{dy}{du} \cdot \frac{du}{dx} = \frac{d}{du}(\ln(u)) \cdot \frac{d}{dx}(\cos x) \\ &= \frac{1}{u} \cdot (-1) \sin x = -\frac{\sin x}{\cos x} = -\tan x \quad (\text{ans}) \end{aligned}$$

(b) let, $u = e^{2x} + 1$ then $y = \sqrt[3]{u}$

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du} \cdot \frac{du}{dx} = \frac{d}{du}(\sqrt[3]{u}) \cdot \frac{d}{dx}(e^{2x} + 1) = \frac{1}{3} \cdot \frac{1}{u^{2/3}} \cdot 2e^{2x} \\ &= \frac{2e^{2x}}{3u^{2/3}} = \frac{2e^{2x}}{3(e^{2x} + 1)^{2/3}} \quad (\text{ans})\end{aligned}$$

(c) let $u = x^5 + 3x^2 + 1$ then, $y = u^{-10}$

$$\begin{aligned}\frac{dy}{dx} &= \frac{dy}{du} \cdot \frac{du}{dx} = \frac{d}{du} u^{-10} \cdot \frac{d}{dx}(x^5 + 3x^2 + 1) = -10u^{-11} (5x^4 + 6x) \\ &= \frac{-10(5x^4 + 6x)}{(x^5 + 3x^2 + 1)^{11}} \quad (\text{ans})\end{aligned}$$

Ans to the question no 16

$$f(x) = y = \cos(\cos x)$$

when, $x = \frac{\pi}{2}$, then, $y = \cos(\cos \frac{\pi}{2}) = 1$

$$\begin{aligned}\text{Here, } m = f'(x) = y_1 &= -\sin(\cos x) \cdot \frac{d}{dx} \cos x = -\sin(\cos x) (-1) \sin x \\ &= \sin(\cos x) \cdot \sin x = \sin(\cos \frac{\pi}{2}) \cdot \sin \frac{\pi}{2} = 0\end{aligned}$$

So, tangent line equation to the graph at $x = \frac{\pi}{2}$ is,

$$(y-1) = 0(x - \frac{\pi}{2})$$

$$\therefore y = 1$$

Ans to the question no 17

$$\begin{aligned}(a) f'(x) &= \frac{d}{dx} \left(\sec^3 \left(\frac{x}{\sqrt{1-x^2}} \right) \right) = 3 \sec^2 \left(\frac{x}{\sqrt{1-x^2}} \right) \frac{d}{dx} \left(\sec \frac{x}{\sqrt{1-x^2}} \right) \\&= 3 \sec^2 \left(\frac{x}{\sqrt{1-x^2}} \right) \sec \left(\frac{x}{\sqrt{1-x^2}} \right) \cdot \tan \left(\frac{x}{\sqrt{1-x^2}} \right) \cdot \frac{d}{dx} \left(\frac{x}{\sqrt{1-x^2}} \right) \\&= 3 \sec^3 \left(\frac{x}{\sqrt{1-x^2}} \right) \cdot \tan \left(\frac{x}{\sqrt{1-x^2}} \right) \cdot \frac{\sqrt{1-x^2} + \frac{2x^2}{2\sqrt{1-x^2}}}{1-x^2} \\&= 3 \sec^3 \left(\frac{x}{\sqrt{1-x^2}} \right) \tan \left(\frac{x}{\sqrt{1-x^2}} \right) \cdot \frac{1 + \frac{x^2}{1-x^2}}{\sqrt{1-x^2}} \\&= \frac{3 \sec^3 \left(\frac{x}{\sqrt{1-x^2}} \right) \tan \left(\frac{x}{\sqrt{1-x^2}} \right)}{(1-x^2) \sqrt{1-x^2}} \quad (\text{ans})\end{aligned}$$

$$\begin{aligned}(b) g'(x) &= \frac{-2 \tan \left(\frac{3}{x} \right)}{(x^7-3)^3} \tan \left(\frac{3}{x} \right) \frac{d}{dx} (x^7-3)^{-2} + (x^7-3)^{-2} \frac{d}{dx} \tan \left(\frac{3}{x} \right) \\&= \frac{-2 \tan \left(\frac{3}{x} \right)}{(x^7-3)^3} \frac{d}{dx} (x^7-3) + \frac{\tan \left(\frac{3}{x} \right) \sec^2 \left(\frac{3}{x} \right)}{(x^7-3)^2} \frac{d}{dx} \left(\frac{3}{x} \right) \\&= \frac{-12x^6 \tan \left(\frac{3}{x} \right)}{(x^7-3)^3} - \frac{3 \sec^2 \left(\frac{3}{x} \right)}{x^2 (x^7-3)^2} \quad (\text{ans})\end{aligned}$$

Ans to the question no 18

Here, $u(x) = f(x)g(x)$

$$\Rightarrow u'(x) = f'(x)g(x) + f(x)g'(x)$$

$$\Rightarrow u'(-3) = f'(-3)g(-3) + f(-3)g'(-3) = 7(-2) + 4(-1) \\ = -18 \quad (\text{ans})$$

Here, $v(x) = \frac{f(x)}{g(x)}$

$$\Rightarrow v'(x) = \frac{f'(x)g(x) - f(x)g'(x)}{\{g(x)\}^2}$$

$$\Rightarrow v'(0) = \frac{f'(0)g(0) - f(0)g'(0)}{\{g(0)\}^2} = \frac{(-1)3 - 7(-2)}{3^2} \\ = \frac{11}{9} \quad (\text{ans})$$

Here, $w(x) = f(g(x))$

$$\Rightarrow w'(x) = f'(g(x)) \cdot g'(x)$$

$$\Rightarrow w'(2) = f'(g(2)) \cdot g'(2) \\ = f'(-3) \cdot 5 = 7 \cdot 5 = 35 \quad (\text{ans})$$

Here, $z(x) = g(x^3 - 2x - 2)$

$$\Rightarrow z'(x) = g'(x^3 - 2x - 2) \frac{d}{dx}(x^3 - 2x - 2) = g'(x^3 - 2x - 2)(3x^2 - 2)$$

$$\Rightarrow z'(2) = g'(2^3 - 2 \cdot 2 - 2)(3 \cdot 2^2 - 2) = g'(2) \times 10 = 50$$

(ans)

Ans to the question no 19

(a) $f(x) = 6 - x; [a, x] = [-2, x]$

Here, Area, $A(x) = \frac{1}{2}(x+2)(6-x+8)$

$$= \frac{1}{2}(x+2)(14-x)$$

$$= \frac{1}{2}(-x^2 - 2x + 14x + 28)$$

$$= \frac{1}{2}(-x^2 + 12x + 28)$$

$$\therefore A'(x) = \frac{1}{2}(-2x + 12)$$

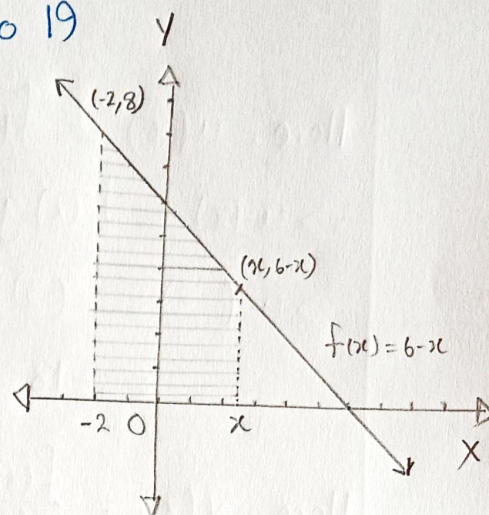
$$= 6 - x$$

$$\therefore A'(x) = f(x) \quad [\text{Confirmed}]$$

Again, $\int_{-2}^x f(x) dx = \int_{-2}^x (6-x) dx = \int_{-2}^x 6 dx - \int_{-2}^x x dx$

$$= (6x + 12) - \left(\frac{x^2}{2} - 2\right) = -\frac{x^2}{2} + 6x + 14$$

$$= \frac{1}{2}(-x^2 + 12x + 28) = A(x) \quad [\text{Verified by direct integration}]$$

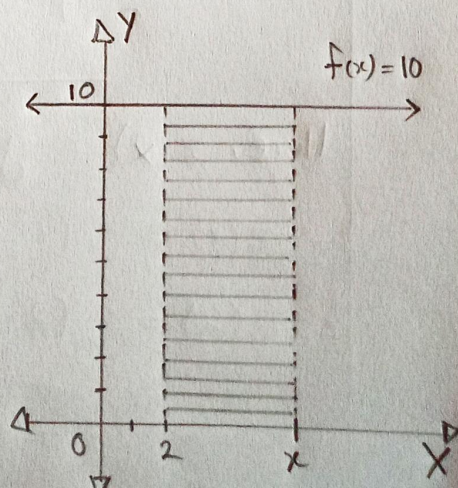


(b) $f(x) = 10; [a, x] = [2, x]$

Here, Area, $A(x) = 10(x-2)$

$$\therefore A'(x) = 10$$

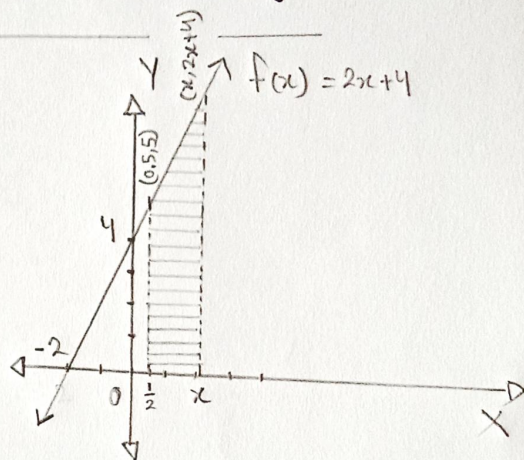
$$\therefore A'(x) = f(x) \quad [\text{Confirmed}]$$



$$\begin{aligned} \text{Again } \int_2^x f(x) dx &= \int_2^x 10 dx = [10x]_2^x = 10x - 20 \\ &= 10(x-2) = A(x) \quad [\text{Verified by direct integration}] \end{aligned}$$

(c) $f(x) = 2x+4$; $[a, x] = [\frac{1}{2}, x]$

$$\begin{aligned} \text{Here, Area, } A(x) &= \frac{1}{2} \left(x - \frac{1}{2}\right) (2x+4+5) \\ &= \frac{1}{2} \left(x - \frac{1}{2}\right) (2x+9) \\ &= \frac{1}{2} \left(2x^2 + 8x - \frac{9}{2}\right) \end{aligned}$$



$$A'(x) = \frac{1}{2} (4x+8) = 2x+4 = f(x) \quad [\text{Confirmed}]$$

$$\begin{aligned} \text{Again, } \int_{1/2}^x f(x) dx &= \int_{1/2}^x (2x+4) dx = \int_{1/2}^x 2x dx + \int_{1/2}^x 4 dx \\ &= \left(x^2 - \frac{1}{4}\right) + (4x-2) = \frac{1}{2} \left(2x^2 + 8x - \frac{9}{2}\right) = A(x) \end{aligned}$$

[Verified by direct integration]